

CASELET, MAXIMA AND MINIMA

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

A survey was conducted among people in five cities – Hyderabad, Chennai, Bengaluru, Kochi and Vizag – about the sports watched by them. The following table gives the number of people, in each city, who watch five sports – Cricket, Football, Hockey, Basketball and Golf. The value alongside each city gives the total number of people surveyed in the city.

City \ Sport	Cricket	Football	Hockey	Basketball	Golf
Hyderabad (112)	48	31	26	33	37
Chennai (132)	59	65	37	25	32
Bengaluru (155)	75	61	58	32	41
Kochi (98)	47	58	32	43	22
Vizag (105)	56	32	29	35	28

Note: All questions pertain to only the people surveyed.

What is the minimum number of the people in Chennai who watch all the five sports?

☐ a) 8

☐ b) 11

☒ c) 25 (Your answer - ✖)

☐ d) 0

Solution:

The total number of instances of people watching sports in Chennai is
 $= 59 + 65 + 37 + 25 + 32 = 218$

As 132 persons were surveyed, the 218 instances can be accounted for by people watching one, two or three sports and no person need to watch all the five sports.

Choice (D)

What is the maximum number of people in Kochi who watch all the five sports?

☐ a) 20

☐ b) 22 (Correct Answer - ✔)

☐ c) 38

☐ d) 40

Solution:

The maximum number of people in Kochi who watch all the five sports is the minimum number of people who watch any one among the given five sports, i.e. 22. Choice (B)

What is the minimum number of people in Bengaluru who watch atmost two of the five sports?

☐ a) 51

☐ b) 58

☒ c) 66 (Correct Answer - ✓)

☐ d) None of these

Solution:

Total number of instances of people watching sports in Bangalore = $75 + 61 + 58 + 32 + 41 = 267$

As 155 people were surveyed, we get the minimum number of people watching atmost two sports when all people who watch sports watch at least three sports.

At most 89 people can watch three sports.

$\therefore$ the minimum number of people watching at most two of the five sports is $155 - 89 = 66$. Choice (C)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

What is the maximum number of people in Hyderabad who watched at least two of the five sports, if it is known that each person watched at least one sport?

Solution:

Total number of instances = $48 + 31 + 26 + 33 + 37 = 175$

As 112 people were surveyed, 112 instances are covered assigning one instance to each person. The remaining 63 instances can be assigned to maximum number of persons if each person is assigned one instance each.

$\therefore$ at most 63 persons could have watched at least two of the five sports. Ans: (63)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

What is the minimum number of people in Vizag who watched at least two of the five sports?

Solution:

Total number of instances = $56 + 32 + 29 + 35 + 28 = 180$

If each person is assigned one instance, a total of 105 instances can be accounted for. The remaining are $180 - 105 = 75$ instances. The number of people can be minimized if we assign the maximum possible number of instances (four, in addition to the one already assigned) to the persons.

$$\therefore \text{number of persons required} = \frac{75}{4} = 19$$

$\therefore$ at least 19 people in Vizag watched at least two of the five sports.

Ans: (19)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

'N' players participated in a tennis tournament played on a knockout basis, i.e., one in which in each round, the winner of a match between two players advance to the next round while the loser is eliminated. The tournament was played in six rounds with byes given in the first round such that there were no byes from the second round onwards. The players are seeded from 1 to N, with player seeded 1 being the top seed, the player seeded 2 being the next and so on till the player seeded N who is the last seed.

Directions for questions 6 and 7: Type in your answer in the input box provided below the question.

What is the minimum value of 'N'?

Solution:

As there are six rounds, the number of players can be any number from 33 to 64. The minimum value of N is 33.

Ans: (33)

Directions for questions 6 and 7: Type in your answer in the input box provided below the question.

What is the maximum number of byes in the first round?

Your Answer: Unattempted

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

The number of byes would be maximum when the value of 'N' is minimum, i.e., 33. As there are no byes from the second round onwards, there would be 31 byes.

Ans: (31)

Directions for questions 8 to 10: Select the correct alternative from the given choices.

An upset is said to happen when a lower seeded player beats a higher seeded player. What is the maximum number of upsets in the tournament?

- ☐ a) N
- ☐ b) $N - 1$ (Correct Answer - ✓)
- ☐ c) $N - 2$
- ☐ d) None of these

Solution:

The number of upsets would be maximum when every match in the tournament is an upset. As there are ' N ' players, the maximum number of upsets would be ' $N - 1$ '.
Choice (B)

Directions for questions 8 to 10: Select the correct alternative from the given choices.

An upset is said to happen when a lower seeded player beats a higher seeded player. What is the maximum number of upsets in the tournament if it is known that the top seed won the tournament and $N = 45$?

- ☐ a) 40
- ☐ b) 39 (Correct Answer - ✓)
- ☐ c) 38
- ☐ d) 43

Solution:

The number of upsets would be maximum when all matches other than those involving seed 1 are upsets. As ' N ' is 45, there would be 44 matches and the maximum number of matches not involving seed 1 would be 39 (when seed 1 is given a bye in the first round).
Choice (B)

Directions for questions 8 to 10: Select the correct alternative from the given choices.

An upset is said to happen when a lower seeded player beats a higher seeded player. What is the minimum number of upsets in the tournament if it is known that the player seeded N won the tournament?

☐ a) 5 (Correct Answer - ✓)

☐ b) 6

☐ c) 8

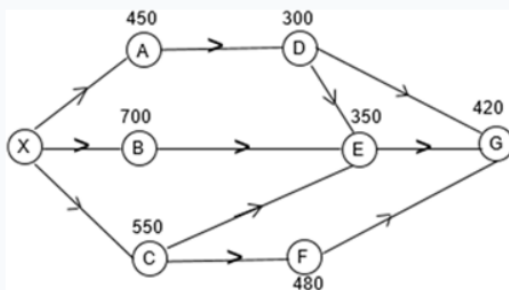
☐ d) 10

Solution:

The minimum number of upsets happen when only matches involving seed N are upsets. If seed N gets a bye in the first round, he/she can win the tournament with five wins, i.e. five upsets.
Choice (A)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The network below gives the supply of oil from a refinery (X) to seven depots, A through G. Oil is transported through pipelines and the arrows represent the direction of flow. The requirement at each depot (in kl) is given alongside it. Oil is transported such that only after the requirement at a depot is met, it is passed on to the next depot in the network. The capacity of all the pipelines connecting the refinery to the depots is 1500 kls and that connecting the depots is 900 kls. Slack in a pipeline is defined as the excess flow required to bring the flow in the pipeline to its maximum capacity.



What is the minimum slack (in kls) in the pipeline connecting D and G?

☐ a) 300

☐ b) 400

☐ c) 450

☐ d) None of these

Solution:

For minimum slack, the flow should be maximum. The maximum flow in the pipeline connecting AD is 900 kls and that in the pipeline connecting DG is 600 kls. The slack is 300 kls.
Choice (A)

What is the maximum slack in the pipeline connecting B and E?

- ☐ a) 750 kls
- ☐ b) 850 kls
- ☐ c) 900 kls
- ☐ d) 800 kls

What is the minimum total flow in the pipelines connected to depot F?

- ☐ a) 520 kls
- ☐ b) 480 kls
- ☐ c) 450 kls
- ☐ d) 400 kls

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

What is the maximum slack (in kls) in all given pipelines?